



Society of Petroleum Engineers

SPE-227285-MS

Optimization of Offshore Production in Real-Time Intelligent Completion Operation with Use of Reservoir-Wellbore Dynamic Coupling Analysis Methods

Zhong Cheng, Liang Zhang, and Zhouzheng Hao, CNOOC Energy Technology & Services Limited Key Laboratory for Offshore Completion, Tianjin, China / CNOOC Ener Tech-Drilling & Production Co., Tianjin, China; Huajie Chu, CNOOC Ener Tech-Drilling & Production Co., Tianjin, China; Chuangang Liu, Shaolin Qu, and Ruifeng Li, CNOOC Energy Technology & Services Limited Key Laboratory for Offshore Completion, Tianjin, China / CNOOC Ener Tech-Drilling & Production Co., Tianjin, China; Zhikun Liu and Ye Liu, Xi'an Shiyu University, Shaanxi, China

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227285-MS](https://doi.org/10.2118/227285-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Oil and gas production efficiency in China's Bohai Bay faces challenges, including difficulties in development and significant issues with high water cut. Furthermore, the rapid development of intelligent offshore oilfields has created an urgent need for remote supervisory control of oil and gas wells, driven by the widespread adoption of unmanned offshore production facilities. To meet the growing demand for intelligent services, we have optimized well completion methods and production management modes and developed a real-time smart well completion control system based on edge-cloud collaborative digital technology. This system integrates intelligent completion downhole control tools and advanced downhole monitoring sensors to monitor near-well reservoir production dynamics in real-time, remotely control oil and gas production, optimize production layer allocation, reduce workover and shutdown frequencies, enhance reservoir and wellbore understanding, mitigate geological heterogeneity impacts, delay gas and water breakthrough, improve production efficiency, extend well life, and enhance offshore oil and gas recovery.

The key to intelligent completion engineering lies in the rational control of inflow control devices to optimize production dynamics and enhance reservoir exploitation management capabilities. Designing and controlling oil and gas field development using smart well completion mainly relies on dynamic simulation and development prediction of well completion and near-well reservoirs. Thus, a sound and reliable simulation scheme, optimal control algorithm, and calculation model are crucial for the successful implementation of smart well completion technology. Using multiphase flow analysis theory in wellbores, coupled with reservoir-well completion data calculation, and wellbore multi-field unsteady coupling analysis technology, we simulate and optimize fluid flow from near-well reservoirs to wellbores. The applied nodal analysis method, known as the vertex-bridge topological structure analysis algorithm, can be expanded radially and axially in the wellbore. It is adaptable to the design and simulation of different

segmentations/layers and applies to multi-segment and multi-branch well completion methods involving inflow control.

This method has undergone on-site testing and application in the S oilfield of Bohai Bay. The results show that inflow control devices in the well have successfully achieved remote control from the production platform to the cloud application. Through coordinated production allocation of multiple zones, no water breakthrough has been observed since production commenced, achieving an overproduction of 10%, thus realizing the objectives of intelligent well production and increased oil recovery.

The advancements presented here offer new strategies for stratified development on offshore platforms, provide a novel approach for production optimization and water control in oil and gas wells, and significantly contribute to the scale-up and operational excellence of China's offshore intelligent oilfields.

Keywords: Intelligent Completion Operation, Dynamic Coupling, Inflow Control Device, Multi-field Unsteady Coupling Analysis, Interval Control Valve

Introduction

As offshore oil and gas fields mature, production systems increasingly face complex challenges such as high water cut, early gas breakthrough, and heterogeneous reservoir behavior. These issues are especially pronounced in offshore environments like China's Bohai Bay, where intervention costs are high and reservoir conditions are geologically complex. In response to these operational challenges, the oil and gas industry has been actively advancing intelligent completion technologies to enhance production efficiency, extend well life, and enable remote, real-time reservoir management.

Among various intelligent completion solutions, ICVs have emerged as a key technology due to their adjustable, remotely controllable, and real-time responsive characteristics. Compared to passive ICDs and AICDs, which operate on fixed flow dynamics and lack adaptability (Ref.1- Ref.2), ICVs enable dynamic regulation of zonal inflow based on real-time downhole data. This functionality allows for timely response to evolving reservoir conditions, facilitating proactive mitigation of water or gas coning, delayed breakthrough, and unbalanced production from different zones (Ref.3- Ref.4).

The integration of ICVs into modern completion systems marks a shift toward closed-loop production optimization, where sensing, simulation, and control form a digital feedback loop. This paradigm typically combines real-time data acquisition via distributed downhole sensors, dynamic multiphase flow simulations, and intelligent control algorithms capable of remotely adjusting valve settings (Ref.5- Ref.6). Recent advancements emphasize the importance of modeling the coupled behavior of reservoir and wellbore systems, including unsteady multi-field interactions and the use of topological analysis methods to capture flow dynamics in stratified formations. Field research in offshore environments further supports this trend. For example, Qilong Zhang et al. presented successful applications of intelligent completions, highlighting challenges related to geological heterogeneity, deployment complexity, and real-time control precision. Their work validates the use of ICVs in enhancing zonal flow control and extending production plateau periods (Ref.7). Similarly, we demonstrated the benefits of integrating edge–cloud collaborative architectures into intelligent well operations, achieving improvements in data transmission efficiency, computational latency, and remote decision-making capabilities (Ref.8). Their findings align with broader research efforts advocating the use of cloud–edge frameworks for real-time optimization (Ref.9).

Despite these advancements, gaps remain in both the modeling and field-scale validation of intelligent completion systems. Most prior work either emphasizes static optimization or relies on laboratory-scale experiments with limited extrapolation to actual offshore production environments. The coupling mechanisms between near-wellbore dynamics and controllable completion components—especially in stratified reservoirs—remain underexplored (Ref.10).

This study aims to address these limitations by designing and field-validating a real-time intelligent completion system based on ICV technology, deployed in a Bohai Bay oilfield. The proposed system

integrates edge-cloud collaborative architecture, dynamic multiphase flow simulation, and remote valve control to achieve fine-scale zonal flow optimization. Field trials evaluate system performance in delaying water breakthrough, balancing multi-zone production, and improving overall recovery. The results contribute to the growing body of knowledge on intelligent well systems and offer a practical framework for scalable implementation in complex offshore environments.

Methodology

In practical field applications, the goal of simulating the coupled reservoir-wellbore system is to evaluate how different ICV settings affect flow distribution and pressure profiles under evolving boundary conditions. As operating conditions change—such as bottomhole pressure, production constraints, or reservoir drive—engineers need to estimate, in real time or in planning scenarios, the resulting flow rates and pressures for a range of ICV opening configurations. These insights are crucial for decision-making processes aimed at flow optimization, water breakthrough control, and balancing production across different zones.

To facilitate these objectives, an intelligent completion system equipped with advanced ICVs offers precise control over fluid inflow into the wellbore from various segments of the reservoir (see Figure 1). This figure provides an overview of the components involved in such a system, including surface control execution equipment, RTU, and ICVs themselves. By dynamically adjusting the openings of these valves based on feedback from downhole sensors, operators can optimize hydrocarbon recovery while minimizing water production and ensuring even drawdown across multiple reservoir layers.

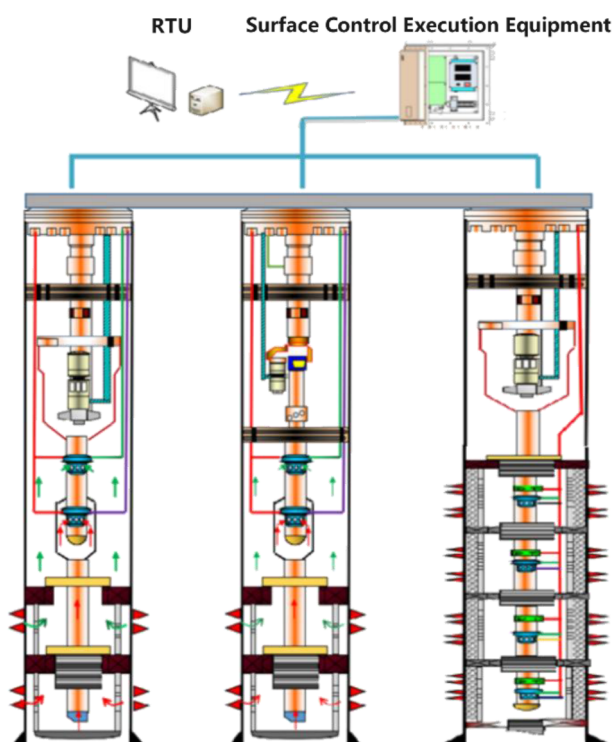


Figure 1—Schematic Diagram of Intelligent Completion with ICV System Components and Flow Control Mechanisms

Numerical Framework for Coupled Reservoir-wellbore Simulation

To evaluate the impact of different ICV settings on flow distribution and pressure profiles, we developed a fully discretized numerical framework that captures the transient, coupled dynamics between the reservoir and wellbore. This model is based on a point-bridge topology, enabling an accurate representation of well-reservoir interactions at discrete interface nodes. Within this framework, segment-wise multiphase flow

in the wellbore is iteratively coupled with the reservoir pressure field, allowing for consistent updates of pressure and flow behavior under arbitrary ICV configurations. The overall computational workflow is illustrated in Figure 2.

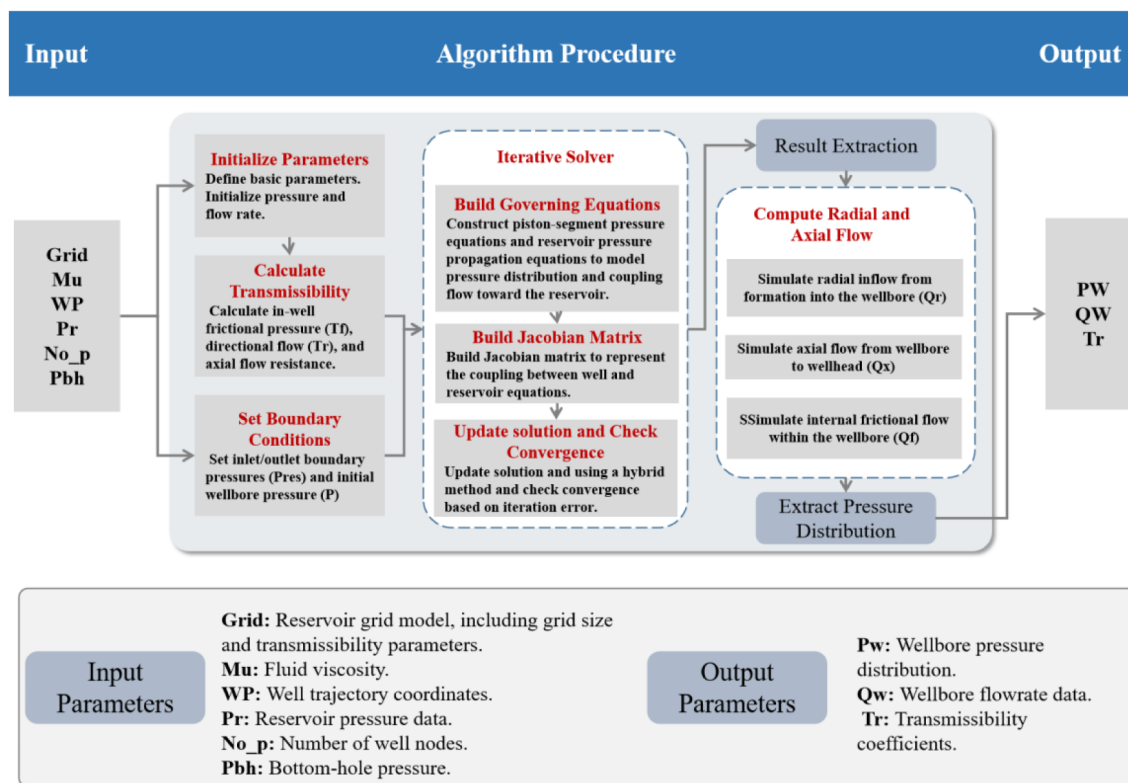


Figure 2—Algorithm Procedure for Coupled Reservoir-Wellbore Simulation

The simulation process begins by specifying key input parameters, including a 3D structured reservoir grid with assigned porosity and permeability values, initial fluid distributions, and phase properties such as density and viscosity. Boundary conditions are defined either by bottomhole pressure or surface production rates. The well trajectory, characterized by measured depth and inclination, is discretized into one-dimensional segments, each representing a control volume. At the interface between the wellbore and reservoir grid, specialized bridge nodes are introduced to capture the bidirectional exchange of mass and pressure between the two domains.

1. **Initialization:** Before time-stepping, transmissibility coefficients are computed to characterize flow conductance throughout the system. These include steady-state formation transmissibility to describe reservoir connectivity and transient transmissibility to capture early-time pressure behavior near the wellbore. Bridge node transmissibility is also calculated, incorporating factors such as skin effects and geometrical deviations near the well interface. Wellbore flow resistance, including gravitational head and frictional losses, is determined based on fluid velocity and trajectory geometry. These parameters form the basis for constructing the Jacobian matrix required for solving the coupled system.
2. **Iterative solver:** At each timestep, an iterative solver is employed to resolve the interdependent pressure fields of the reservoir and wellbore. Initially, given the ICV valve openings and the current wellbore pressure estimate, the reservoir domain is solved using transmissibility-based finite-volume discretization. The resulting pressure distribution provides an updated driving force for calculating bridge-node inflows. The flow rates derived from bridge nodes are then applied as source terms

in a segmented multiphase flow model along the wellbore. Pressure gradients are computed using correlations that incorporate effects from fluid composition, friction, gravity, and inclination angle.

3. **Convergence check:** Following the update of wellbore pressure, the solver checks for convergence by evaluating changes in the flow rates at bridge nodes between successive iterations. If the iteration error is below a predefined threshold, the solution is considered converged, and the simulation proceeds to the next timestep. Otherwise, the updated pressure values are fed back into the reservoir solver, and the coupling loop continues.
4. **Result extraction:** Upon convergence at each timestep, the model outputs include updated reservoir pressure and saturation distributions, detailed wellbore flow profiles (including pressure, velocity, and phase fractions), flow rates through each ICV segment, and local performance indicators such as pressure drop and throttling efficiency. These outputs provide the foundation for performance evaluation, operational adjustment, and production forecasting under various ICV control strategies.

Tool Description and Experimental Characterization

To support intelligent inflow management in horizontal wells, our team has independently developed an ICV tool based on a hydraulic actuation design. The tool integrates three main functional modules: displacement control, hydraulic piston, and aperture regulation (see Figure 3). The displacement control unit receives hydraulic signals from the surface and converts them into mechanical motion. The hydraulic piston translates this displacement into axial movement, which in turn adjusts the aperture control mechanism to modulate the flow area through the valve. This modular structure ensures precise and repeatable control over valve openings under downhole pressure and temperature conditions.

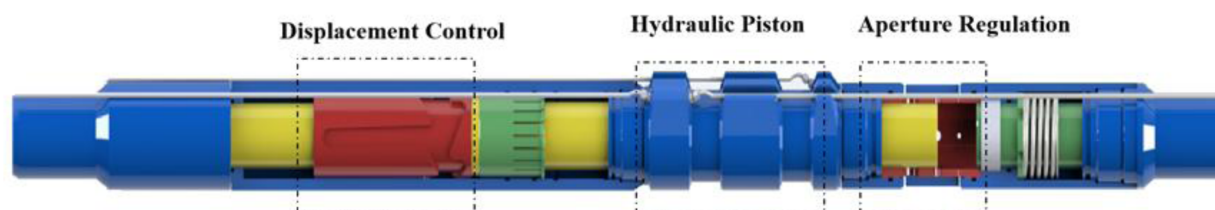


Figure 3—Schematic Diagram of Self-Developed Equipment with Key Functional Modules

To evaluate the flow regulation performance of the tool, we conducted a dedicated HTHP experimental study to measure the valve's flow coefficient (C_v) under realistic operating conditions. As shown in Figure 4, the tool was subjected to a series of controlled C_v tests, where flow rates and pressure drops were recorded at various aperture settings.



Figure 4—High-Temperature and High-Pressure Testing of ICV Equipment in a Laboratory Setting

The resulting C_v data demonstrates a nonlinear response across the full range of valve openings (see Figure 5). Notably, the measured C_v values deviate significantly from theoretical expectations derived from API standards-API specification 19ICV 2023, especially under partially open conditions where complex flow behaviors such as turbulence and contraction dominate. This discrepancy highlights the importance of direct experimental characterization: relying solely on standard coefficients would result in substantial errors when modeling ICV-reservoir interactions.

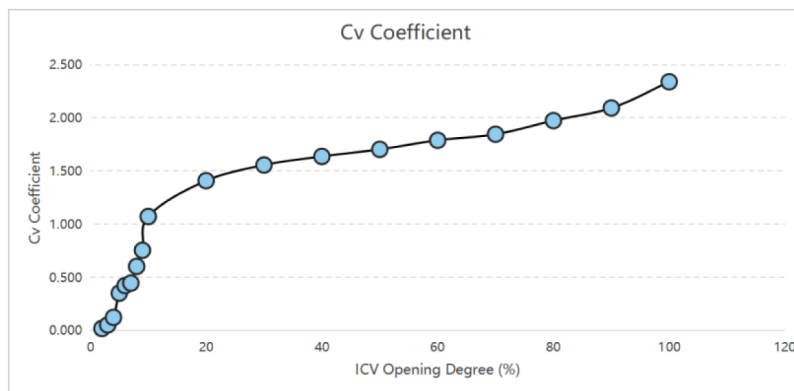


Figure 5—Relationship Between ICV Opening Degree and C_v Coefficient

Overall, this experiment provides critical empirical support for the integration of the ICV tool into numerical simulations. The experimentally derived C_v profiles offer a more realistic and reliable foundation for inflow control modeling, enabling accurate prediction of segment-wise pressure and flow behavior under various valve settings.

The experimental results reveal significant deviations from standard C_v estimates based on API guidelines, particularly in partially open positions where turbulence and flow contraction effects dominate. This finding underscores the importance of direct tool-specific calibration: relying solely on generic standards can lead to substantial modeling errors in flow prediction and reservoir coupling analysis. Therefore, incorporating empirically measured C_v values into the numerical simulator enhances both the fidelity and reliability of our ICV-integrated production forecasting framework.

Experiment

To assess the applicability of the proposed simulation framework in real-field conditions, a horizontal production well (Well 1) from a field operated by CNOOC was selected for detailed analysis. The well is located within a target block and features a multi-segment completion with integrated ICVs, making it a representative case for evaluating dynamic flow control strategies.

As illustrated in Figure 6, the horizontal section of Well 1 extends through the reservoir. The two ICVs are installed at depths of 2797 m and 3055 m, respectively, dividing the completion into two distinct control intervals. Both segments are equipped with six-stage hydraulically actuated ICVs, enabling precise regulation of inflow.

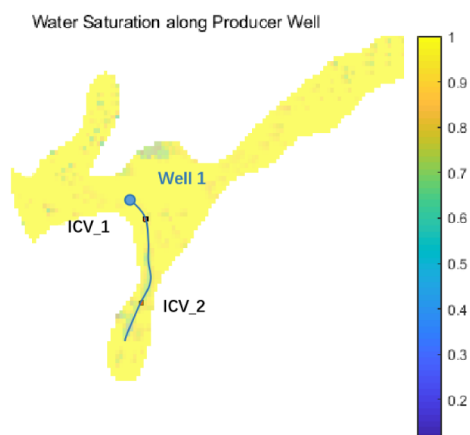


Figure 6—Plan View of Water Saturation Distribution and Simulated Well Trajectory with ICV Locations

Simulation-Based Optimization for Well 1

To explore the potential of ICV-based flow control optimization, a simulation and optimization study was conducted on a representative well, Well 1, using the open-source reservoir simulation platform MRST. The goal was to optimize the valve control settings of ICVs to maximize oil recovery under realistic geological and operational constraints.

A reservoir model was established based on the real geological model from target block, simplified to a single-layer grid. Geological structure and petrophysical properties were updated using an external GRDECL file, which defines porosity, permeability, and other key attributes of the target block. A black-oil fluid model was adopted, specifying relevant physical parameters such as oil and water viscosities and densities, along with Corey-type relative permeability curves.

The well configuration included one injection well and two producer wells, each placed at designated grid locations. The two producer wells were equipped with ICVs, allowing for dynamic control of segmental flow. The simulation schedule spanned 36 control intervals, with valve settings permitted to change every 60 days. Initially, a forward simulation was performed under baseline valve settings to generate a reference case. Subsequently, the optimization process began by introducing connection transmissibility multipliers as control parameters, effectively representing the degree of ICV openness. These control variables were grouped by well and parameterized across time steps.

The optimization objective was to maximize NPV, accounting for oil revenue and water handling costs (both production and injection). The NPV function was defined to reflect realistic economic considerations over the project duration. A gradient-based L-BFGS optimization algorithm was used to iteratively update the valve control parameters. Sensitivity information was efficiently computed via adjoint methods, enabling scalable evaluation of gradients with respect to control variables.

Figure 7 presents the optimization process, illustrating the evolution of the objective function, gradient norm, line search iterations, and scaled control parameters over iterations. The results demonstrate stable convergence and meaningful control updates. Figure 8 displays the final optimized ICV configurations across 14 control steps. Notably, ICV_1 exhibits a progressive increase in openness, while ICV_2 remains largely unchanged, suggesting spatial differences in reservoir contribution.

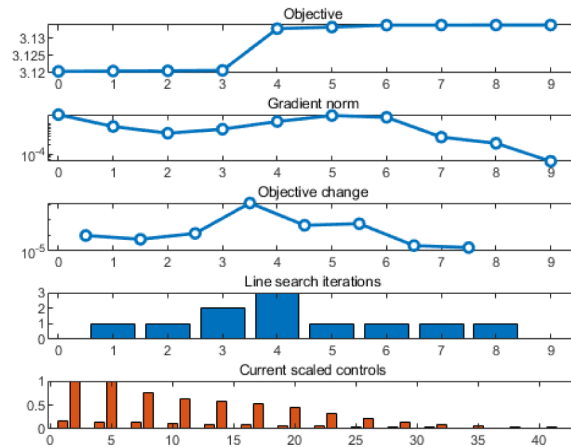


Figure 7—Optimization process tracking for valve control parameters using L-BFGS algorithm

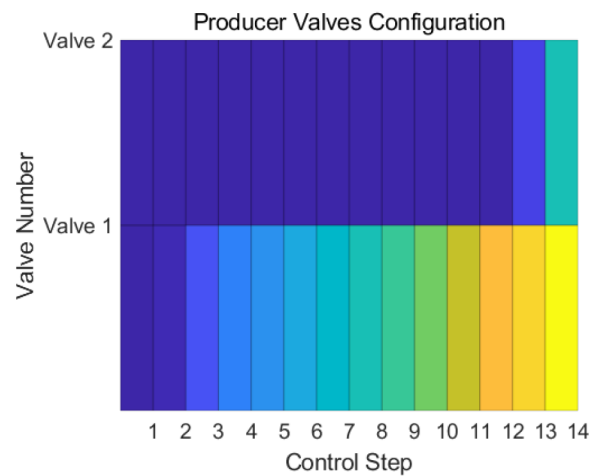


Figure 8—Optimized ICV configurations for Well 1 over control steps

Simulation Optimization Analysis

The combined analysis of oil and water production curves under baseline and optimized ICV control scenarios provides a comprehensive understanding of the optimization's impact on reservoir performance. As depicted in Figure 9 and Figure 10, which illustrate the temporal evolution of oil and water production rates for Well 1 respectively, the effectiveness of dynamic ICV adjustments becomes evident. Both figures share a common time axis ranging from 0 to 2500 days, facilitating a direct comparison between the baseline (blue curve) and optimized (red dashed curve) scenarios.

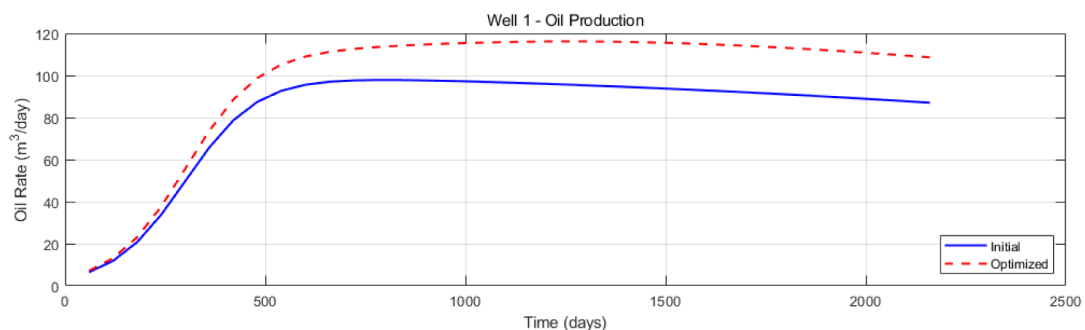


Figure 9—Comparison of Oil Production Rates Over Time Between Scenarios Without and With Optimized ICV Control for Well 1

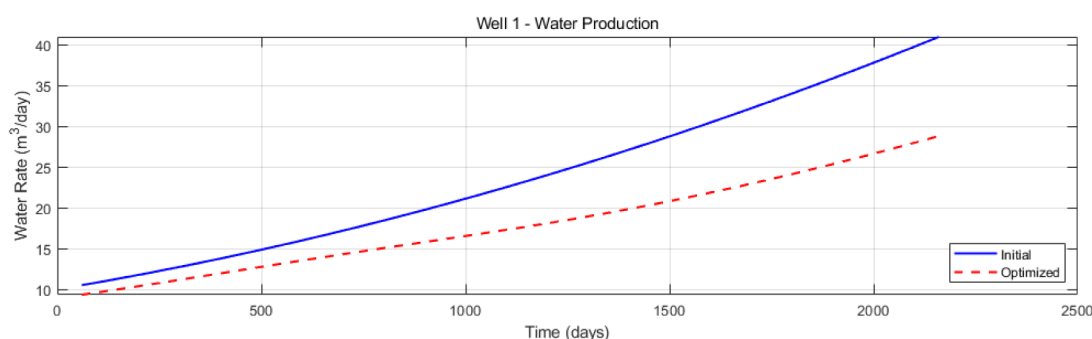


Figure 10—Comparison of Water Production Rates Over Time Between Scenarios Without and With Optimized ICV Control for Well 1

In the context of oil production, the baseline scenario shows an initial rapid increase in production rates, reaching approximately 90 m³/day by day 300 before stabilizing. In contrast, the optimized scenario demonstrates a more pronounced increase, achieving a peak production rate of around 115 m³/day within the same timeframe. This improvement can be attributed to the staged opening pattern of ICV_1, which allows for better access to high-permeability zones while delaying pressure decline. The result is a sustained higher oil output throughout the simulation period, indicating enhanced reservoir management through precise flow control.

Simultaneously, the water production analysis reveals a significant reduction in water influx when comparing the baseline and optimized scenarios. While the baseline case experiences a steady rise in water production, ultimately reaching about 40 m³/day by the end of the simulation period, the optimized scenario exhibits a markedly slower increase, stabilizing at roughly 30 m³/day. This reduction underscores the capability of dynamic ICV settings to delay water breakthrough, particularly by minimizing the early influx from high-permeability segments. Such control strategies not only enhance oil recovery but also mitigate the operational costs associated with water handling and disposal.

The synergistic effects observed in both oil and water production curves highlight the trade-offs involved in optimizing reservoir performance. By maximizing oil recovery while simultaneously reducing water production, the proposed ICV control strategy effectively balances economic efficiency and environmental sustainability. These findings further validate the robustness of the model, demonstrating its potential application in managing complex well architectures such as those encountered in Well 1. Overall, this integrated analysis offers valuable insights into the benefits of incorporating advanced data-driven control methods into reservoir management practices.

Conclusion

This study presented a data-driven approach for optimizing ICV settings in segmented horizontal wells, with the objective of enhancing oil recovery while minimizing water production. The methodology was evaluated using a representative simulation model based on a real well in Bohai bay. The results demonstrated that dynamic adjustment of ICVs significantly improves reservoir performance by enabling precise control over inflow from different segments, particularly in heterogeneous formations.

The simulation-based optimization framework, implemented using the MRST platform, successfully identified optimal valve configurations that maximized NPV over the production period. Adjoint-based sensitivity analysis enabled efficient gradient computation, facilitating stable convergence of the L-BFGS optimization algorithm. The optimized ICV settings led to a notable increase in oil production-reaching up to 115 m³/day during peak periods-while simultaneously reducing water cut compared to the baseline scenario without active flow control. Moreover, the comparative analysis of production curves clearly illustrated the benefits of incorporating real-time ICV adjustments into reservoir management strategies. Delayed water breakthrough and sustained oil rates were observed in the optimized case, highlighting the effectiveness

of segmental flow control in mitigating early coning and channeling effects. These findings underscore the economic and environmental advantages of intelligent well completion technologies in modern oilfield development.

Acknowledgement

The authors wish to thank and acknowledge CNOOC management for permission to publish this paper.

Nomenclature

ICV	= Interval Control Valve
ICD	= Inflow Control Device
AICD	= Autonomous Inflow Control Device
RTU	= Remote Terminal Unit
3D	= Three Dimensions
HTHP	= High Temperature and High Pressure
C_v	= Fow Relationship (US gpm/psi ^{0.5})
API	= American Petroleum Institute
CNOOC	= China National Offshore Oil Corporation
MRST	= Matlab Reservoir Simulation Toolbox
GRDECL	= A File Format for Describing Geological Models
NPV	= Net Present Value
L-BFGS	= Limited-memory Broyden-Fletcher-Goldfarb-Shanno

References

1. Giuseppe A. Rosi, Da Zhu, Ian D. Gates, Jingyi Wang. Passive Flow Control Devices—Well Design and Physics of Their Different Flow Regimes: A Review. *presented at "Journal of Petroleum Science and Engineering"* 2022. **218**: p.110999.
2. Mahmoud Abd El-Fattah, Ahmed Moustafa Fahmy, Hamed Wahaibi, Abdullah Shibli, Khaled Zuhaimi. Autonomous Inflow Control Valve Applications Creating Better Wells: A Review of Design Optimization and Associated Benefits. Paper IPTC21411 presented at the International Petroleum Technology Conference, Virtual, March 2021.
3. Reza Malakooti, Ahmad Zhafran Ayop, Belladonna Maulianda, Khafiz Muradov, David Davies. Integrated Production Optimisation and Monitoring of Multi-zone Intelligent Wells. *presented at "Journal of Petroleum Exploration and Production Technology"* 2020. **10**: p.159–170.
4. Rahim Masoudi, Keng Seng Chan, Hooman Karkooti, Sumit Soni, Shlok Jalan, Noman Shahreyar, Tejas Kalyani, Douglas Finley. Intelligent Completions Customized to Meet the Challenges of Malaysia Reservoirs: Work Flow and Case Histories. Paper SPE175522 presented at the SPE Offshore Europe Conference and Exhibition, Aberdeen, Scotland, UK, September 2015.
5. Zhi Zhang, Baojiang Sun, Zhiyuan Wang, Shaowei Pan, Wenqiang Lou, Dalin Sun. Intelligent Well Killing Control Method Driven by Coupling Multiphase Flow Simulation and Real-time Data. *presented at "Journal of Petroleum Science and Engineering"* 2022. **213**: p.110337.
6. Jing Wan, Bruce A. Dale; Tim K. Ellison, Tim G. Benish, Marcel A. Grubert. Coupled Well and Reservoir Simulation Models to Optimize Completion Design and Operations for Subsurface Control. Paper SPE113635 presented at the Europec/EAGE Conference and Exhibition, Rome, Italy, June 2008.

7. Qilong Zhang, Zhong Li, Haoyang Weng, Wei Liu, Zhongwei Huang and Bin Zhang. Research and Application of Intelligent Well Completion Technology in Offshore Oilfield. *presented at "Journal of Physics: Conference Series"* 2024. 2834.
8. Zhong Cheng, Liang Zhang, Zhouzheng Hao, Shaolin Qu, Jingchao Liu, Xiangxiang Ding, Chuangang Liu, Ning Li, Ruifeng Li. Assisting Offshore Production Optimization Technology in Intelligent Completion Operation Based on Edge-Cloud Collaborative Technologies. Paper SPE222157 presented at the ADIPEC, Abu Dhabi, UAE, November 2024.
9. Md Emran Hossain, Md Tanvir Rahman Tarafder, Nisher Ahmed, Abdullah Al Noman, Md Imran Sarkar, Zakir Hossain. Integrating AI with Edge Computing and Cloud Services for Real-time Data Processing and Decision Making. *presented at "International Journal of Multidisciplinary Sciences and Arts"*, 2023. **2**(4): p.252–261.
10. Fuziana Tusimin, Latief Riyanto, Nurul Aula A'akif Fadzil, Nur Syazana Sadan, Asba Mazidah Abu Bakar, Nik Fazril Ain Sopian, A. Hafriz A. Wahid, M. Farris Bakar, Noman Shahreyar, Zulkiflee Hamzah. Achieving Injectivity Optimization and Zonal Rates Allocation through Multi-zone Intelligent Completion Technology-A Field B Case Study from Offshore Malaysia. Paper SPE201044 presented at the IADC/SPE Asia Pacific Drilling Technology Conference, Virtual, June 2021.